Data Science and Artificial Intelligence

Machine Learning

Linear Discriminant Analysis

Lecture No. 3

By- SIDDHARTH SABHARWAL SIR

Recap of Previous Lecture

- Topic LDA'S
- Topic
- Topic
- Topic
- Topic

Topics to be Covered

Topic LDA Question

Topic Advantage- disadvantage

Topic PCA

Topic

Topic

LIFE IS LIKE
a camera
JUST FOCUS
ON WHAT IS IMPORTANT
CAPTURE
THE GOOD TIMES
DEVELOP
FROM THE NEGATIVES
IF THINGS DON'T WORK OUT
take another shot!

REVISION

LDA $\Rightarrow$

$$\left(\max \frac{w^T S_B w}{w^T S_W w}\right) = J(w)$$

$$\left(\underbrace{S_W^{-1} S_B}_{d \times d \text{ matrix}} \; w = \frac{A}{B} w\right)$$

evectors

evalue

REVISION

LDA

- Step 1 $S_W^{-1} S_B$
- find E. values $|S_W^{-1} S_B - \lambda I| = 0$
- $[S_W^{-1} S_B - \lambda I][w] = 0$ find eigen vectors
- we get d eigenvalues $\Rightarrow$ To convert into K dimension we take 'K' largest eigen values

$$\begin{bmatrix} w \end{bmatrix}_{d \times K}$$

Linear Discriminant Analysis

Practise

1. What is the primary goal of Linear Discriminant Analysis (LDA)?

a) To maximize the variance of the data. → PCA

b) To minimize the covariance between classes.

c) To maximize the separability between different classes.

d) To find non-linear boundaries between classes.

Practise

2. Which assumption is NOT made by LDA?

a) The features follow a Gaussian distribution.

b) The covariance matrices of the classes are equal.

c) The relationships between the features are linear.

d) The features are independent of each other.

Practise

3. In LDA, the Within-Class Scatter Matrix (S_W) measures:

a) The variance within each class.

b) The variance between the classes.

c) The covariance of the entire dataset.

d) The overall mean vector of the data.

$$S_W = S_1^2 + S_0^2$$
$$= (X_1 - \mu_1)(X_1 - \mu_1)^T + (X_0 - \mu_0)(X_0 - \mu_0)^T$$

Practise

4. How is the Between-Class Scatter Matrix (S_B) computed in LDA?

$S_B = (\mu_1 - \mu_0)(\mu_1 - \mu_0)^T$

a) By summing the covariance matrices of all classes.

b) By summing the outer product of the difference between the class means and the overall mean.

c) By calculating the mean of all data points. ✓

d) By multiplying the within-class scatter matrix by its transpose.

Practise

5. Which of the following is a key difference between LDA and PCA?

a) LDA maximizes variance while PCA maximizes class separability.

b) PCA maximizes variance while LDA maximizes class separability.

c) Both LDA and PCA maximize class separability.

d) Both LDA and PCA maximize variance.

Linear Discriminant Analysis

Practise

6. In the context of LDA, what is the primary purpose of eigenvectors and eigenvalues?

a) ~~To find the~~ principal components of the data.

b) To transform the data into a new space with reduced dimensions.

c) To maximize the covariance within each class.

d) ~~To minimize the between-class scatter.~~

b Variance

LDA → within class ~~~ min

b/w class ~~~ max

Seperability.

Practise

7. Which step comes first in the LDA algorithm?

a) Compute the eigenvectors and eigenvalues.

b) Transform the samples to the new space.

c) Compute the mean vectors of each class.

d) Sort the eigenvectors by decreasing eigenvalues.

Practise

8. LDA is particularly effective when:

a) The classes are linearly separable.

b) The classes are not linearly separable.

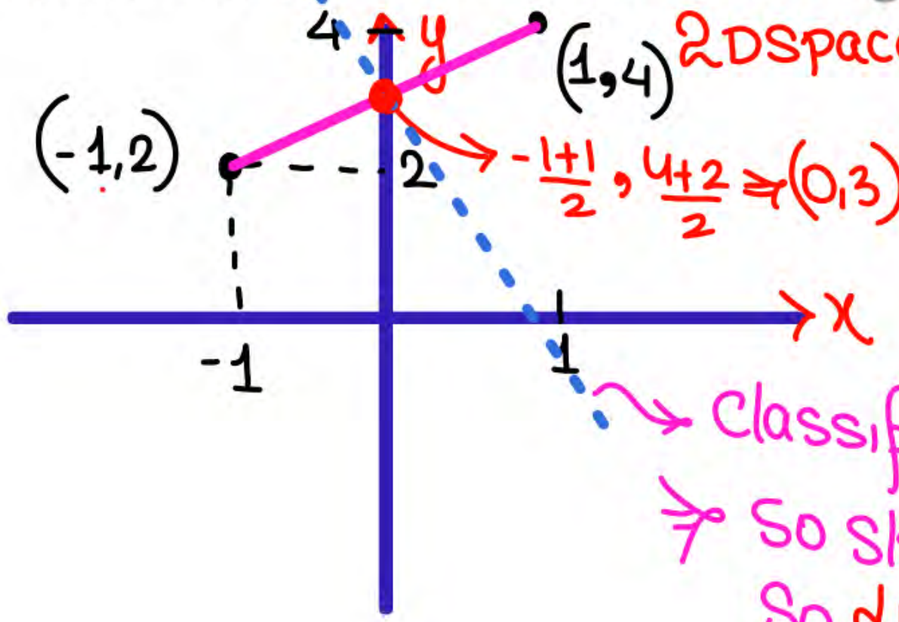
c) There are more features than samples.

d) The data follows a non-Gaussian distribution.

NPTEL

Practise

5. Consider the case where two classes follow Gaussian distribution which are centered at (−1, 2) and (1, 4) and have identity covariance matrix. Which of the following is the separating decision boundary?



2D space

MAP Rule

$\rightarrow \frac{-1+1}{2}, \frac{4+2}{2} \Rightarrow (0,3)$

Classifier ⊥ to Joining line

$\Rightarrow$ So slope = −1

So line $\Rightarrow y = -x + 3$.

(a) $y - x = 3$

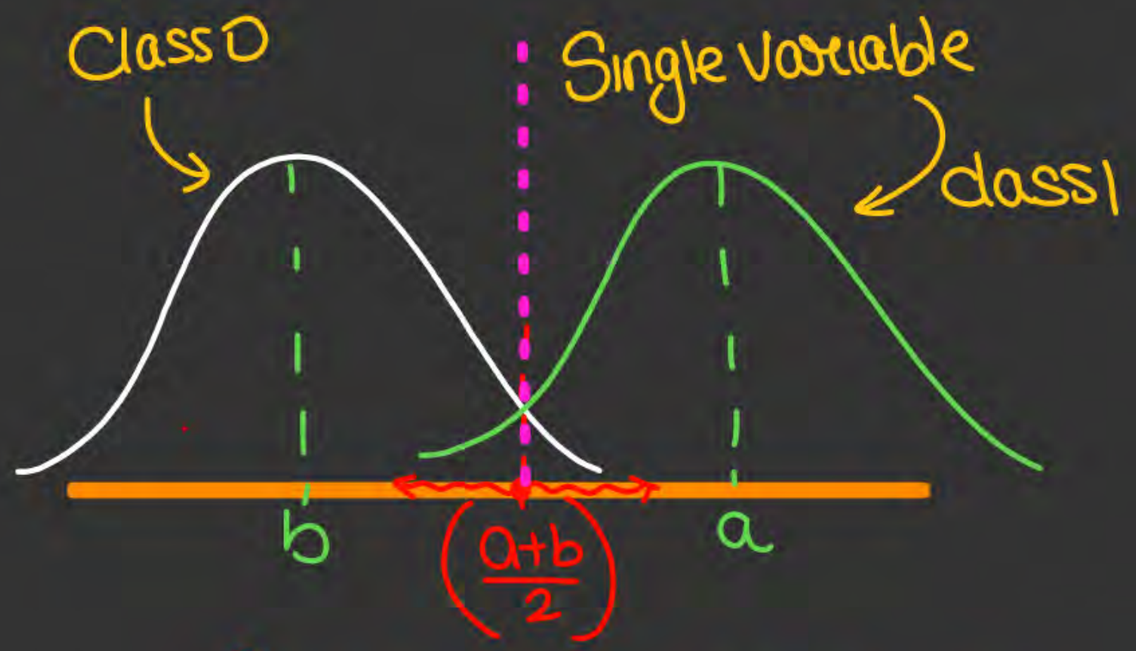
(b) $x + y = 3$

(c) $x + y = 6$

(d) (b) and (c) are possible

e) None of these

(f) Can not be found from the given information



If any test point $x > \frac{a+b}{2}$ class 1

$x < \frac{a+b}{2}$ class 0

Features ⇒ Naive Bayes

8. We discussed the use of MLE for the estimation of parameters of logistic regression model. We used which of the following assumptions to derive the likelihood function ?

$P(x_1, x_2, x_3 \ldots / w, b) \Rightarrow$
$= P(x_1 / w, b) P(x_2 / w, b) \ldots$

a. independence among the class labels ✓

b. independence among each training sample

c. independence among the parameters of the model

w, b ✓

d. None of these

1. What is the primary goal of Linear Discriminant Analysis (LDA)?

a. Dimensionality reduction

b. Feature selection

c. Classification

d. Clustering

3. Which of the following is an assumption of LDA?

a. Features are normally distributed

b. Features have equal variances

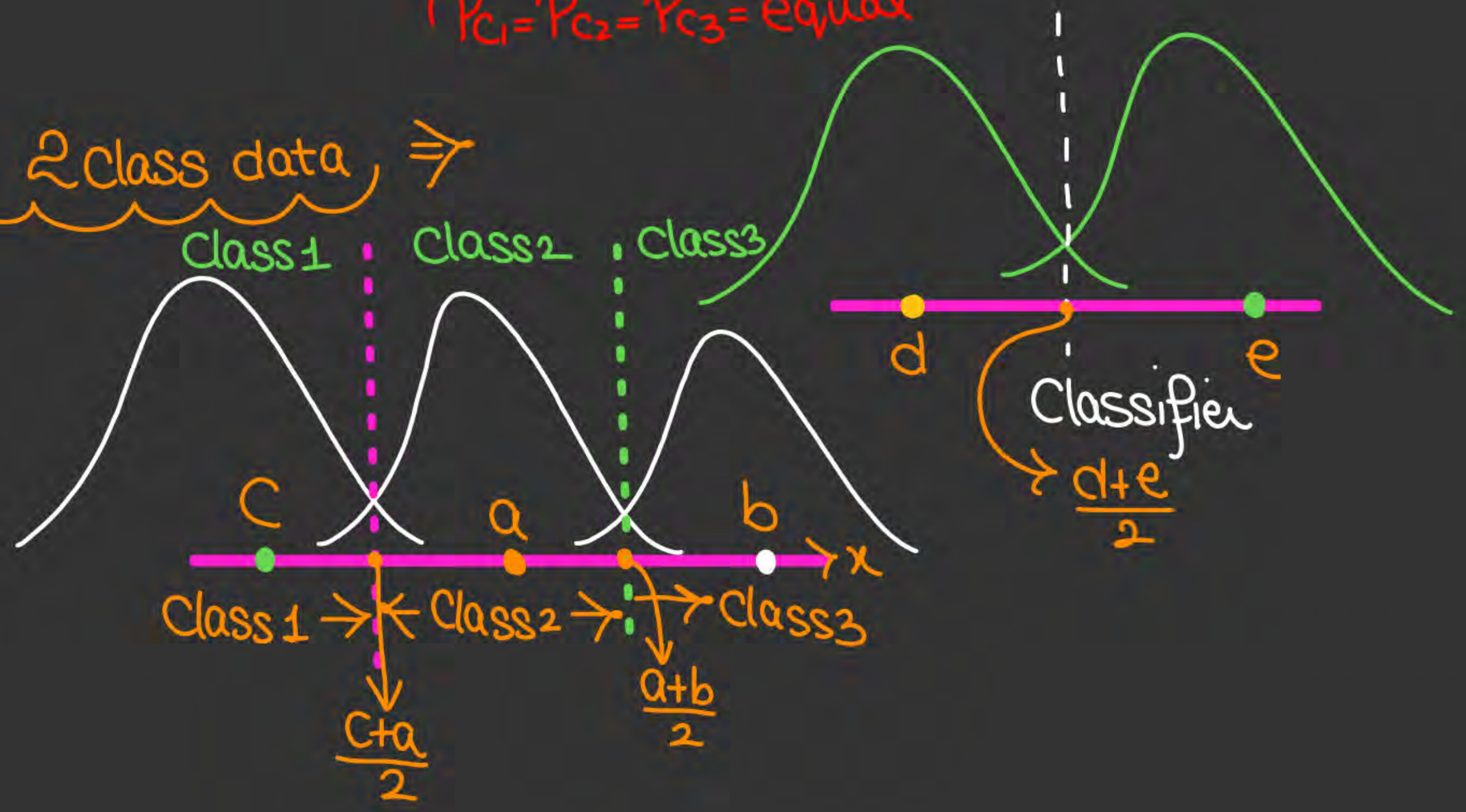
c. Features are independent

d. Features are sparse

Prior

$P_{C_1} = P_{C_2} = P_{C_3} =$ equal

2 class data ⇒



5. How many linear discriminants can you have in LDA for a classification problem with 'k' classes?

a. k

b. k - 1

c. k + 1

d. k^2

6. In LDA, what is the within-class scatter matrix?

a. A measure of the spread of data points within each class

b. A measure of the spread of data points across all classes

c. A measure of the covariance between classes

d. A measure of the variance within a single class

$$\max \frac{(\tilde{\mu}_1 - \tilde{\mu}_0)^2}{s_0^2 + s_1^2}$$

7. Which value represents a good separation between classes in LDA?

a. Small between-class scatter and large within-class scatter

b. Small within-class scatter and large between-class scatter

c. Equal between-class and within-class scatter

d. No scatter at all

9. What is the value of Fisher's criterion (J) for perfect class separation in LDA?

max

a. J = 0

b. J = 1

c. J = ∞

d. J = -1

10. LDA assumes that classes have a Gaussian distribution. What is the assumption regarding the covariance matrices of different classes?

a. They are always the same.

b. They are different, but not necessarily diagonal.

c. They are diagonal.

d. They are not considered in LDA.

- So we have Projection in lower dimension
- $\tilde{\mu}_1, \tilde{\mu}_0$

$$P_{C_1} = \frac{\text{No of points of class } C_1}{\text{Total No of Points}}$$

13. What is the LDA decision rule for classifying a new data point to one of the classes?

a. Choose the class with the highest posterior probability.

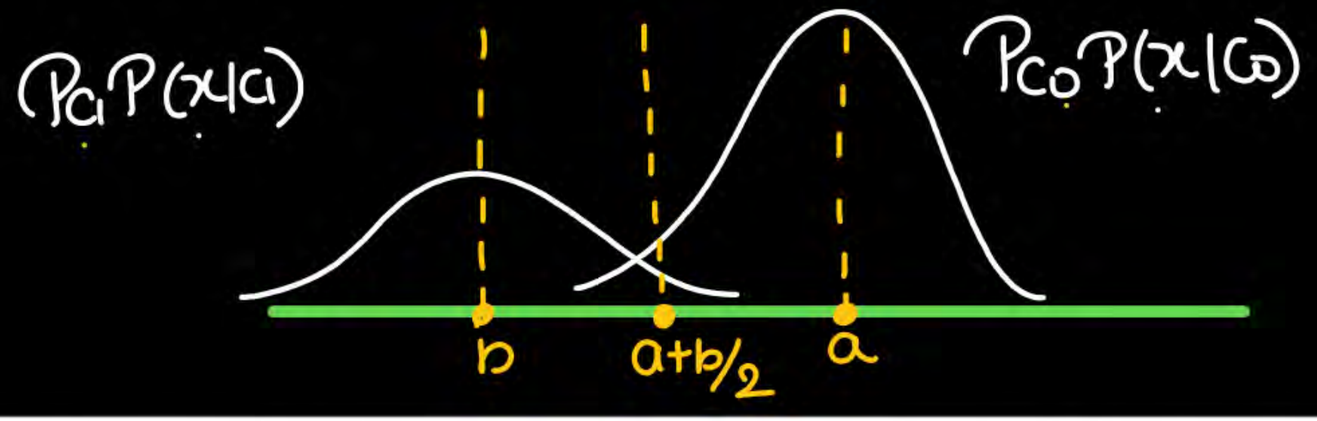
b. ~~Choose the class with the lowest prior probability.~~

c. Choose the class with the smallest discriminant score.

d. ~~Choose the class with the largest covariance.~~

MAP Rule

$$P_{C_1} P(x/C_1) \gtrless P_{C_0} P(x/C_0)$$



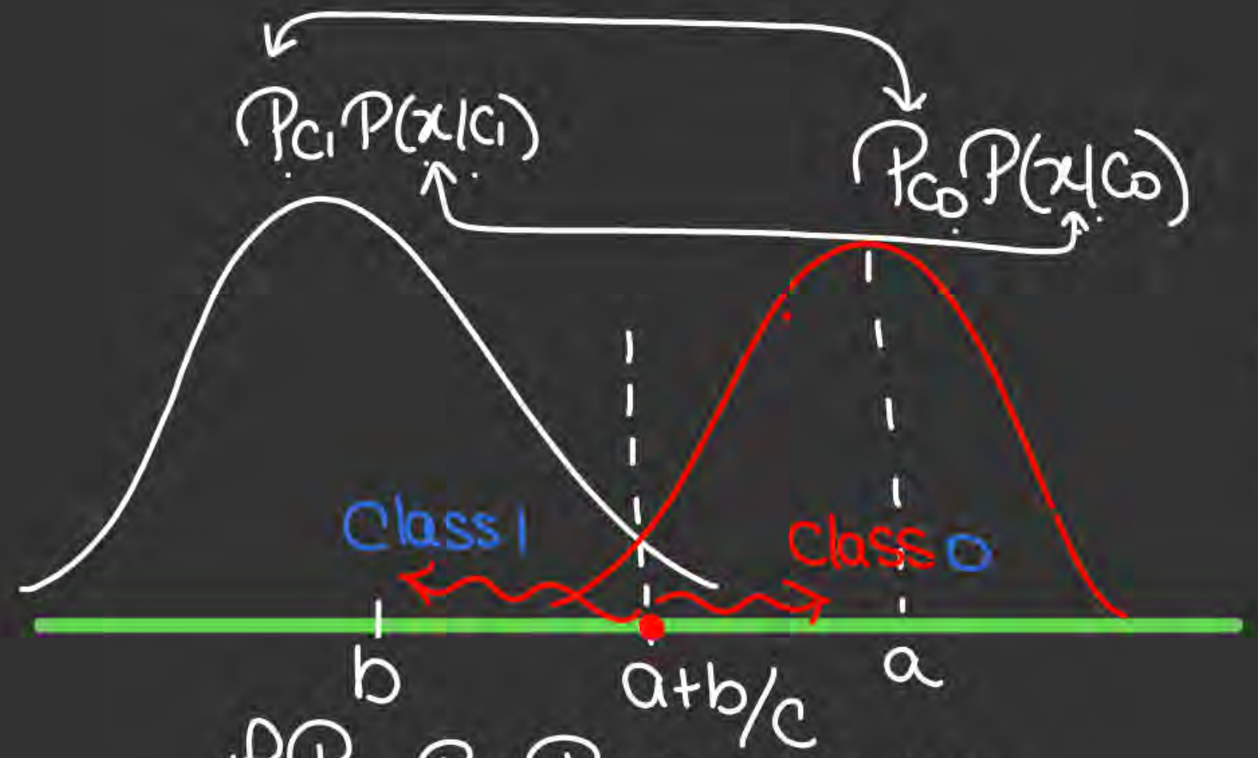
15. In LDA, what is the effect of class priors on the decision boundary?

a. ~~Class priors have no effect on the decision boundary.~~

b. ~~Higher priors shift the decision boundary towards that class.~~

c. Lower priors shift the decision boundary towards that class. (c)

d. Class priors make the decision boundary curved.



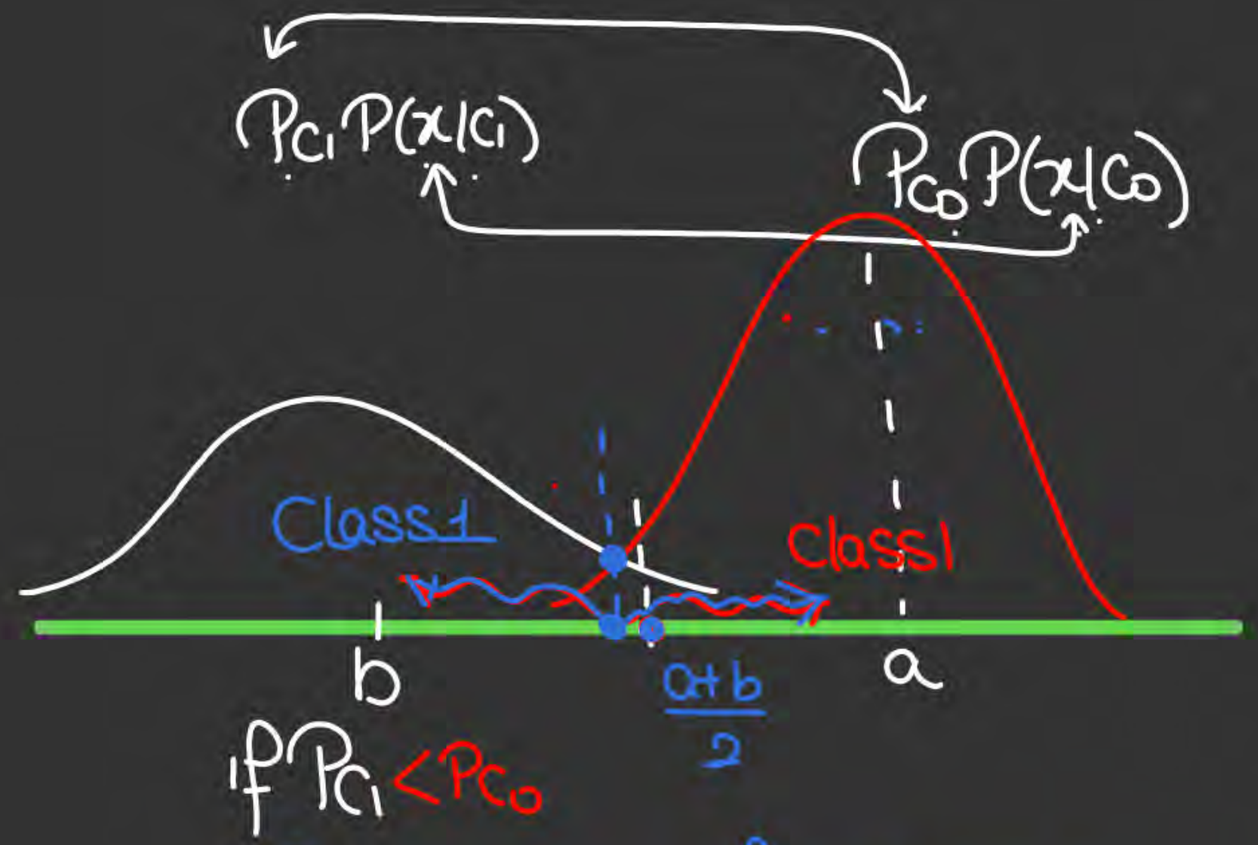
If $P_{C_1} = P_{C_0}$ Prior are same

So Posterior $P_{C_0} P(x|C_0)$

$P_{C_1} P(x|C_1)$

Since in LDA we assume that the distribution in Gaussian and with in class Covariance is Same

- $P(x|C_1) = P(x|C_0)$
- Only mean diff



if $P_{C_1} < P_{C_0}$

Prior reduce threshold shift toward it

Since in LDA we assume that the distribution in Gaussian and with in class Covariance is Same

- $P(x|C_1) = P(x|C_0)$
- Only mean diff

dimensions

20. You are performing LDA on a dataset with 5 ~~classes~~. How many eigenvalues and eigenvectors are typically obtained when you calculate the discriminant directions?

a. 5 eigenvalues and 1 eigenvector

b. 4 eigenvalues and 4 eigenvectors

c. 5 eigenvalues and 5 eigenvectors

d. 4 eigenvalues and 1 eigenvector

dxd

24. If the within-class scatter matrix (Sw) is a 4x4 matrix and the between-class scatter matrix (Sb) is a 4x4 matrix, what is the dimension of the weight vector (W) in LDA?

a. 1x4

b. 4x4

c. 4x1

d. 1x1

$[\quad]_{d\times 1}$

Advantages & Disadvantages of using LDA

- Advantages of using LDA
- It is a simple and computationally efficient algorithm.
- It can work well even when the number of features is much larger than the number of training samples.
- It can handle multicollinearity (correlation between features) in the data.

Advantages & Disadvantages of using LDA

- Disadvantages of LDA
- It assumes that the data has a Gaussian distribution, which may not always be the case.
- It assumes that the covariance matrices of the different classes are equal, which may not be true in some datasets.
- It assumes that the data is linearly separable, which may not be the case for some datasets.

• Within class covariance ⇒ Same ⇒

means ⇒

So {dataset} → class 0 → Class 0 data

→ Class 1 → Class 1 data

Cov(X, Y) ⇒

X and Y are two RV

N points

$$\frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

$Cov(X, X) \Rightarrow \sigma_x^2$

Class 0 data

$$
\begin{array}{c|cccc}
x^1 & x_1^1 & x_2^1 & & x_N^1 \\
x^2 & x_1^2 & x_2^2 & & x_N^2 \\
x^3 & x_1^3 & x_2^3 & & \vdots \\
\vdots & \vdots & \vdots & & \vdots \\
x^D & x_1^D & x_2^D & & x_N^D
\end{array}
$$

So Covariance matrix

$$
\begin{bmatrix}
C_{x_1x_1} & C_{x_1x_2} & C_{x_1x_3} & & C_{x_1x_D} \\
C_{x_2x_1} & C_{x_2x_2} & C_{x_2x_3} & - & - \\
| & | & | & & | \\
| & | & | & & |
\end{bmatrix}
$$

$D \times D$

So Multivariate Gaussian.

if we have $x^1\ x^2\ x^3 \ldots x^D$ dimensions

then the distribution of Class 0/Class 1 points

$$\Rightarrow \frac{1}{(2\pi)^{D/2}\,|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}[X-\mu]^T(\Sigma^{-1})[X-\mu]\right)$$

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_D \end{bmatrix}$$

$\Rightarrow$ ($\Sigma \Rightarrow$ Covariance matrix)

$S_W^{-1} S_B$ → Symmetric matrix

→ eigenvectors.

So it means the reduced dimensions are orthogonal to each other

Some Important Facts...

1 Some Facts on Symmetric Matrices

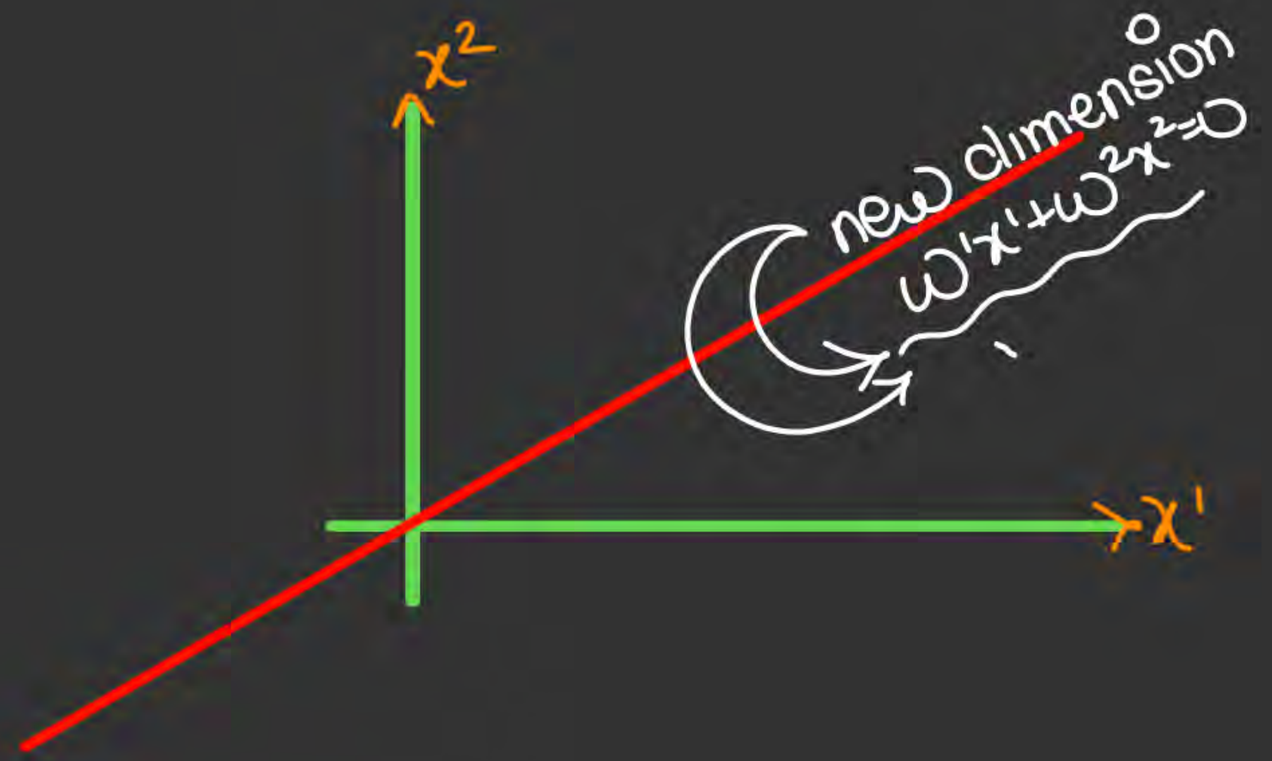
Definition: Matrix A is symmetric if $A = A^T$.

Theorem: Any symmetric matrix

1) has only real eigenvalues;

2) is always diagonalizable;

3) has orthogonal eigenvectors.


x^2
x^1
new dimension
$w^1x^1+w^2x^2=0$

PCA ⇒ Unsupervised learning technique

→ means y label of data is not needed

→ whole data is analysed together, such that we find the new dimensions ⇒ called principal components.

Such that the variation / variance / information is correctly captured onto the lower dimensions.

- Tries to max. the variance of the collected data

What is PCA

- Principal component analysis (PCA) is a dimensionality reduction and machine learning method used to simplify a large data set into a smaller set while still maintaining significant patterns and trends.
- This is a unsupervised learning method.
- Principal Component Analysis (PCA) is a statistical procedure that uses an orthogonal transformation that converts a set of correlated variables to a set of uncorrelated variables. ← Becoz new features are ortho. → original features
- Principal Component Analysis (PCA) is used to reduce the dimensionality of a data set by finding a new set of variables, smaller than the original set of variables, retaining most of the sample's information, and useful for the regression and classification of data.

LDA

new dimension ⇒ linear Combination of old dimension

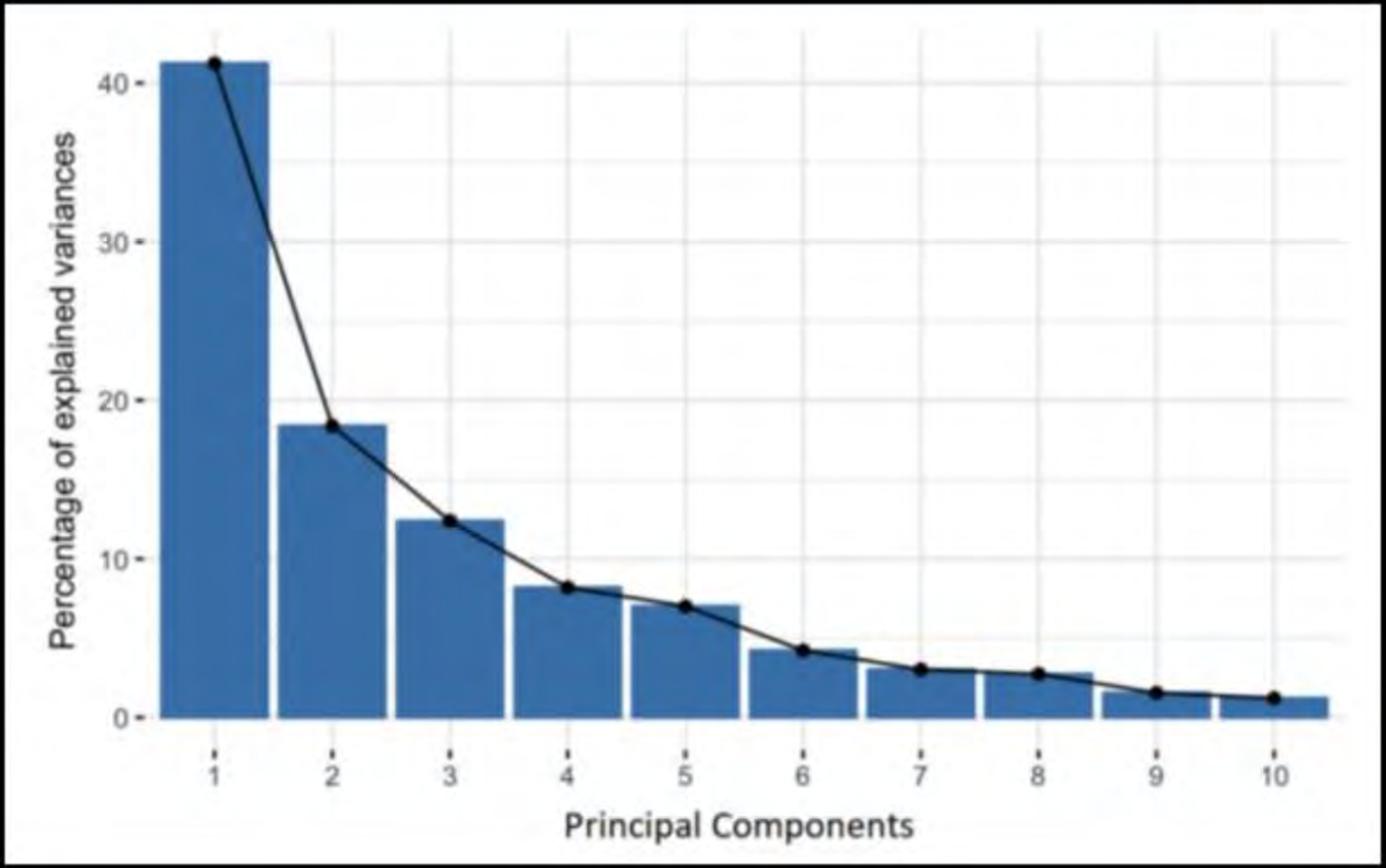
What is PCA

- Principal Component Analysis (PCA) is a technique for dimensionality reduction that identifies a set of orthogonal axes, called principal components, that capture the maximum variance in the data. The principal components are linear combinations of the original variables in the dataset and are ordered in decreasing order of importance. The total variance captured by all the principal components is equal to the total variance in the original dataset.
- The first principal component captures the most variation in the data, but the second principal component captures the maximum variance that is orthogonal to the first principal component, and so on.

What is PCA

- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.
- These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.
- So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on, until having something like shown in the scree plot below.

What is PCA


Percentage of explained variances
40
30
20
10
0
1
2
3
4
5
6
7
8
9
10
Principal Components

What is PCA

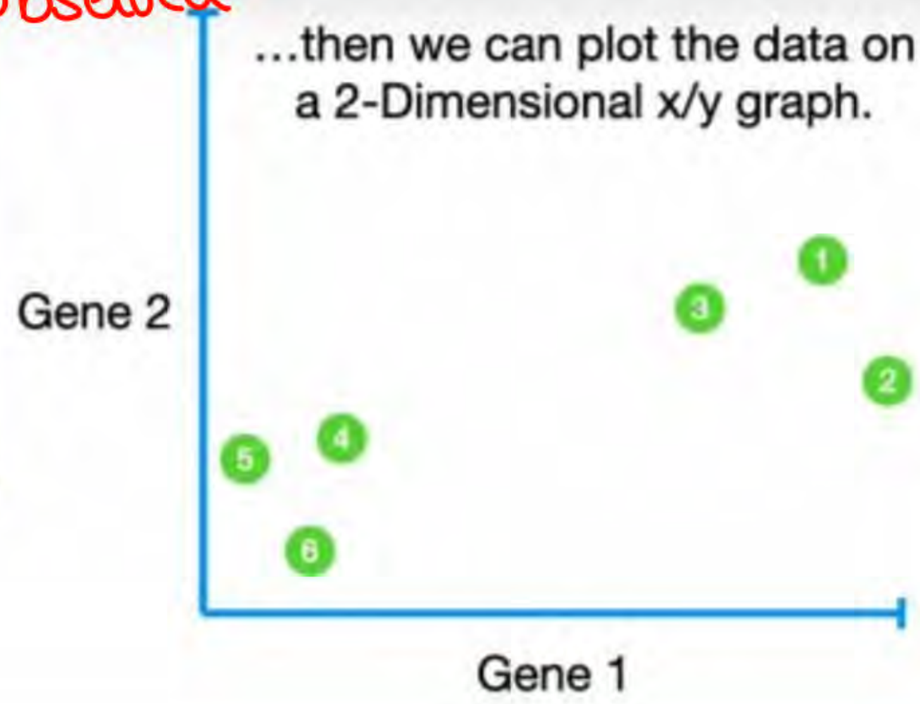
We can see that points 1,2,3 and 4,5,6, make clusters

no class, whole data is obsewed

Samples

	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	1	2
Gene 2	6	4	5	3	2.8	1

...then we can plot the data on a 2-Dimensional x/y graph.

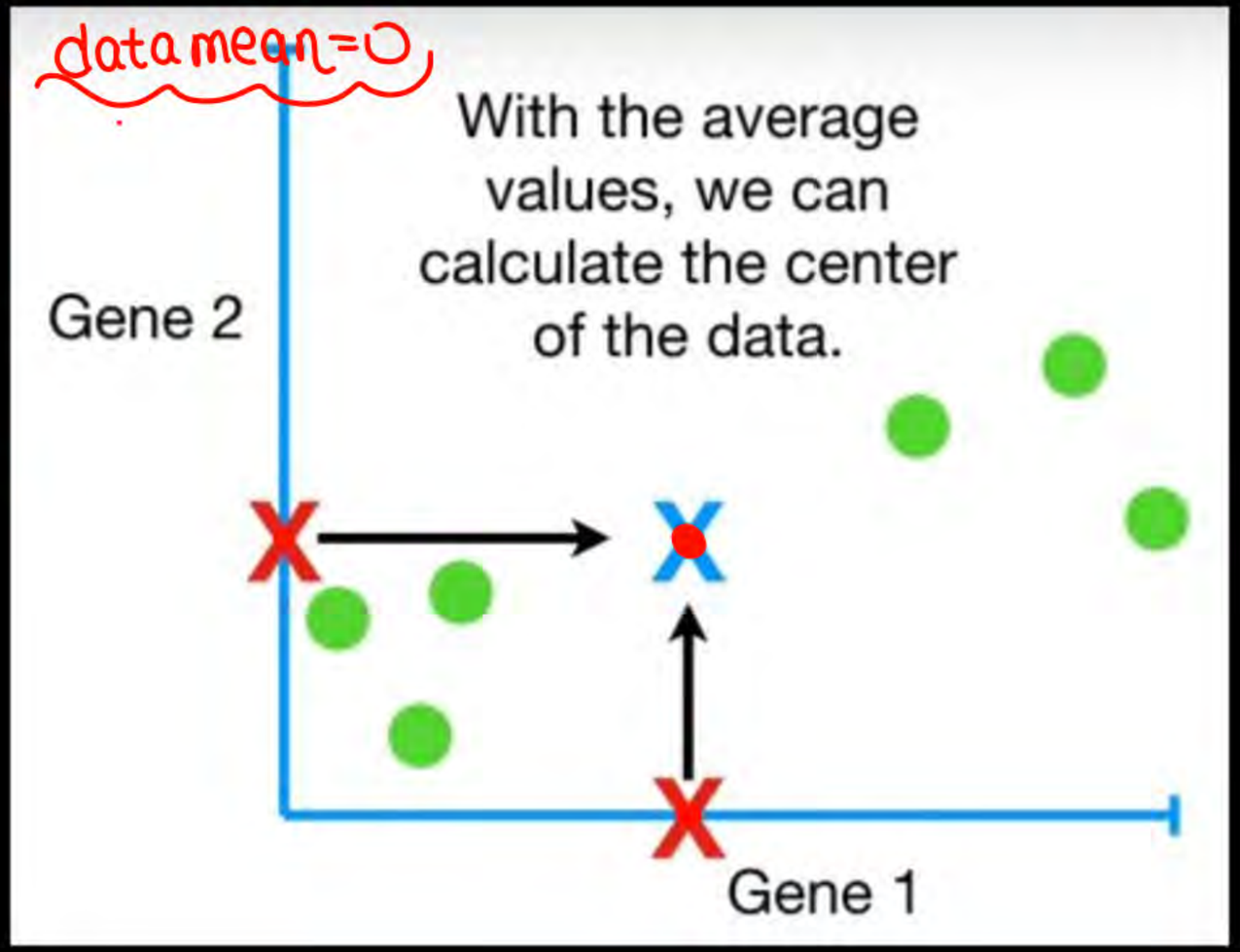


What is PCA

First of all we calculate the centre of the data

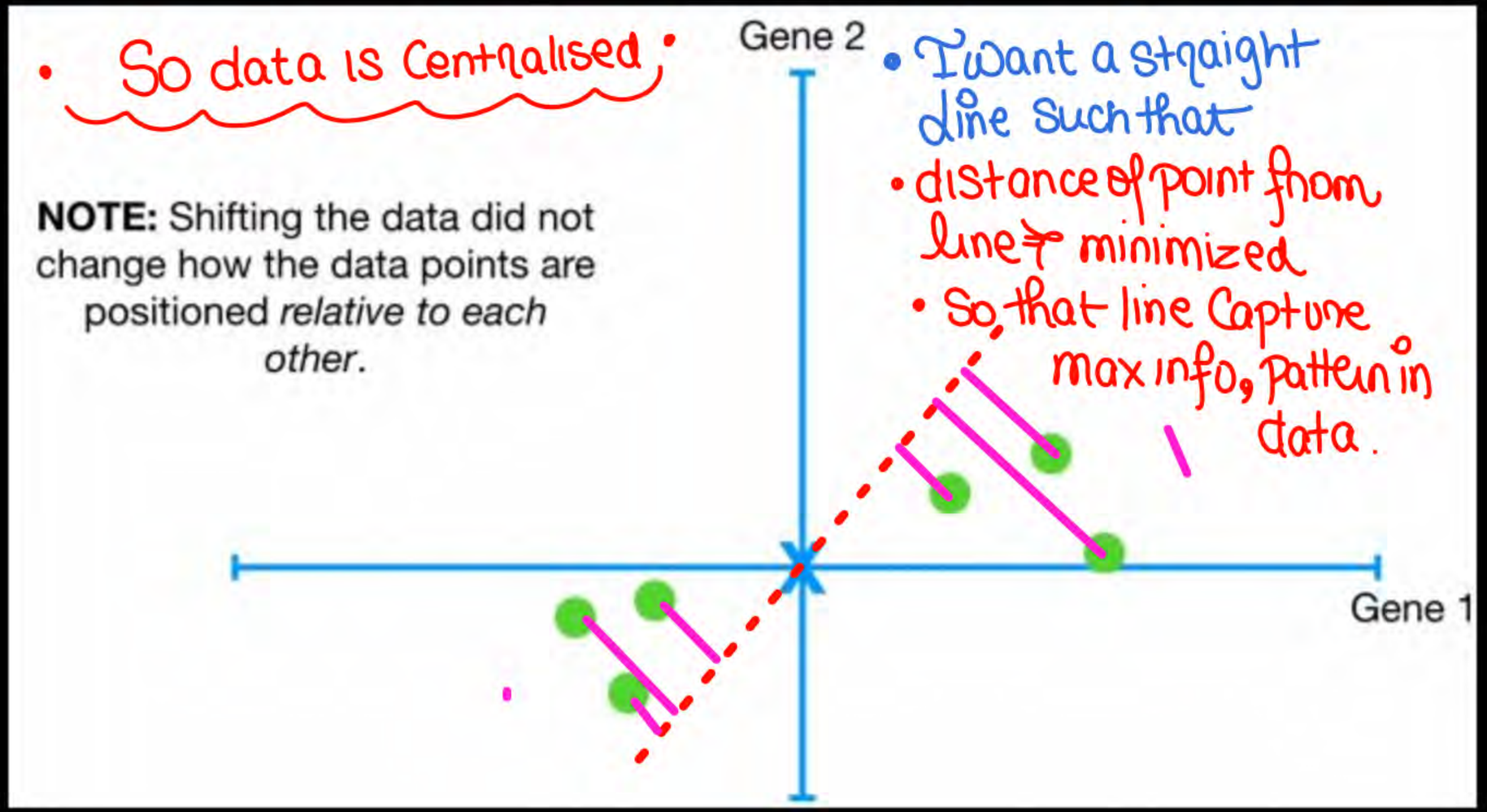
$$X \Rightarrow \mu = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_d \end{bmatrix}$$

$$[X-u]$$



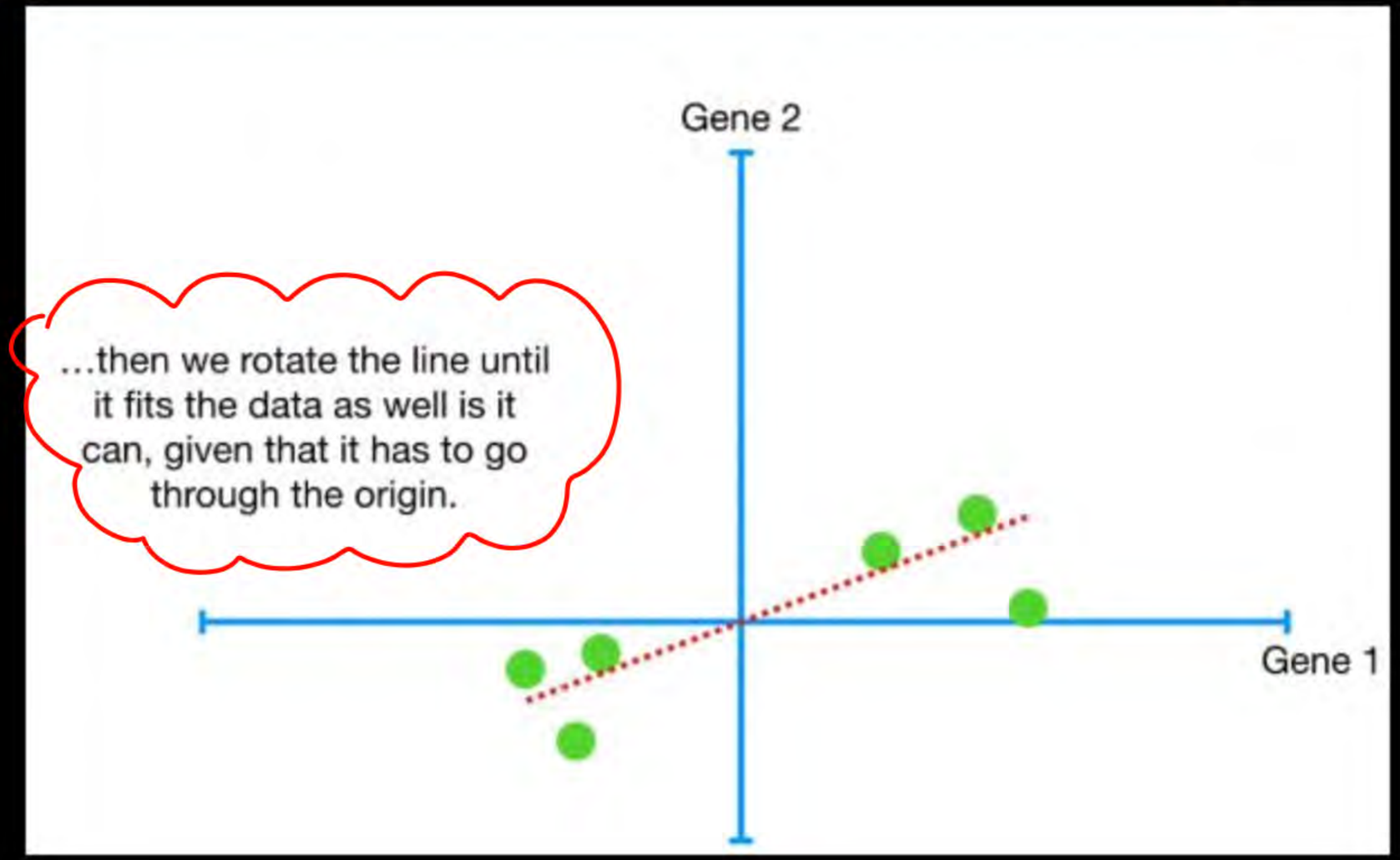
What is PCA

We shift the date to make its centre at the origin….

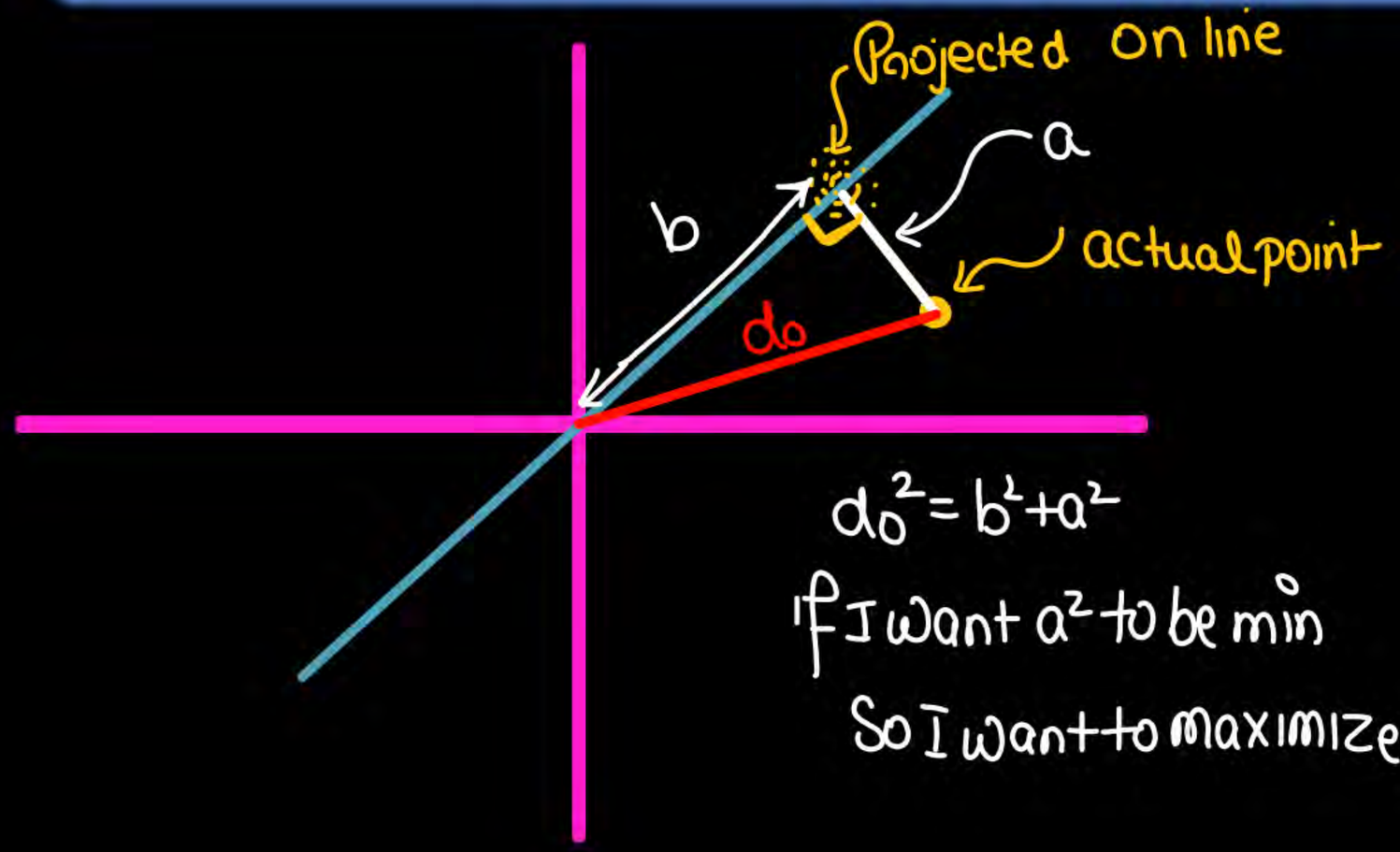


What is PCA

Now the PCA try to fit a line to the data…
Thus we need a line which has the minimum distance of point from the line or….



What is PCA



$d_o^2 = b^2 + a^2$

If I want a^2 to be min

So I want to maximize b^2

• d_o = fix, becoz distance of point from origin is fix

So I can say that the best fit line is that where the distance of Projection of Point from Origin is max.

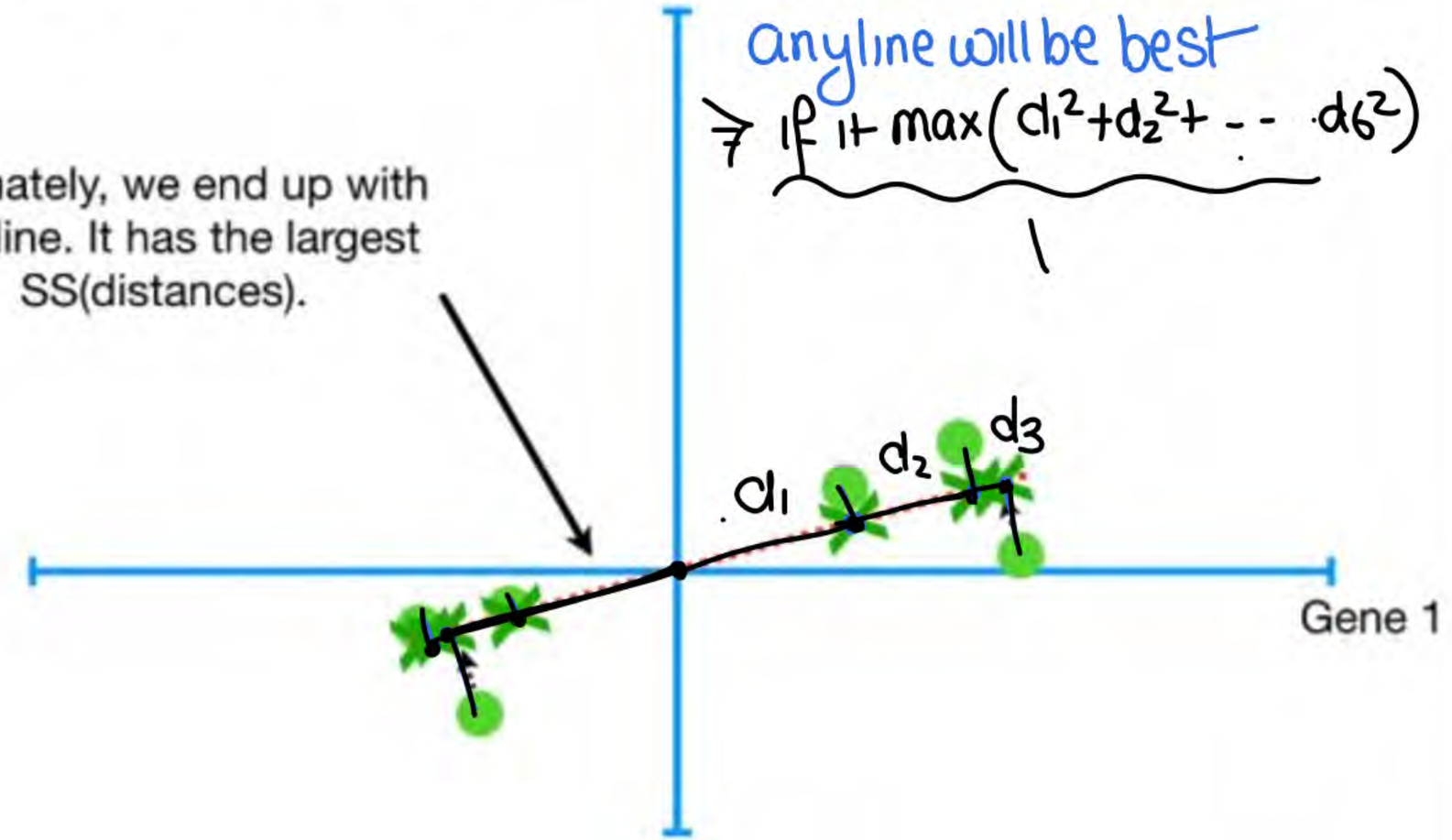
What is PCA

$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

anyline will be best

$\Rightarrow$ if it max$(d_1^2 + d_2^2 + \cdots d_6^2)$

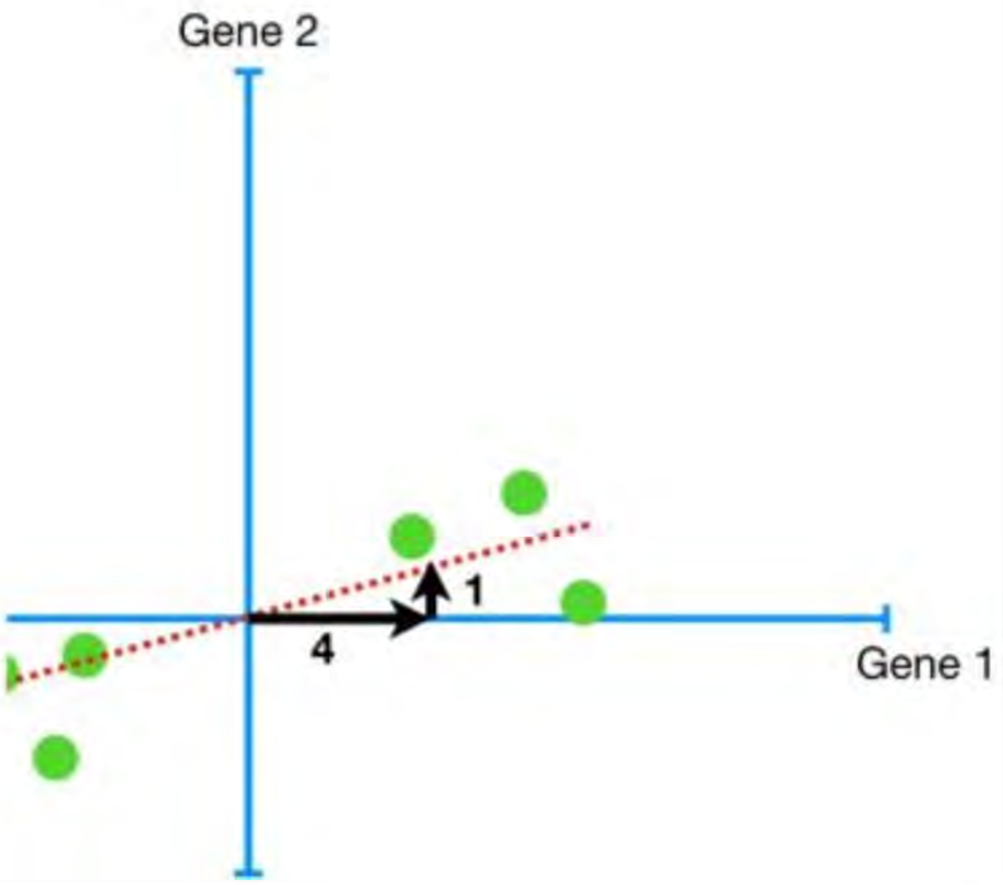
Ultimately, we end up with this line. It has the largest SS(distances).



What is PCA

To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

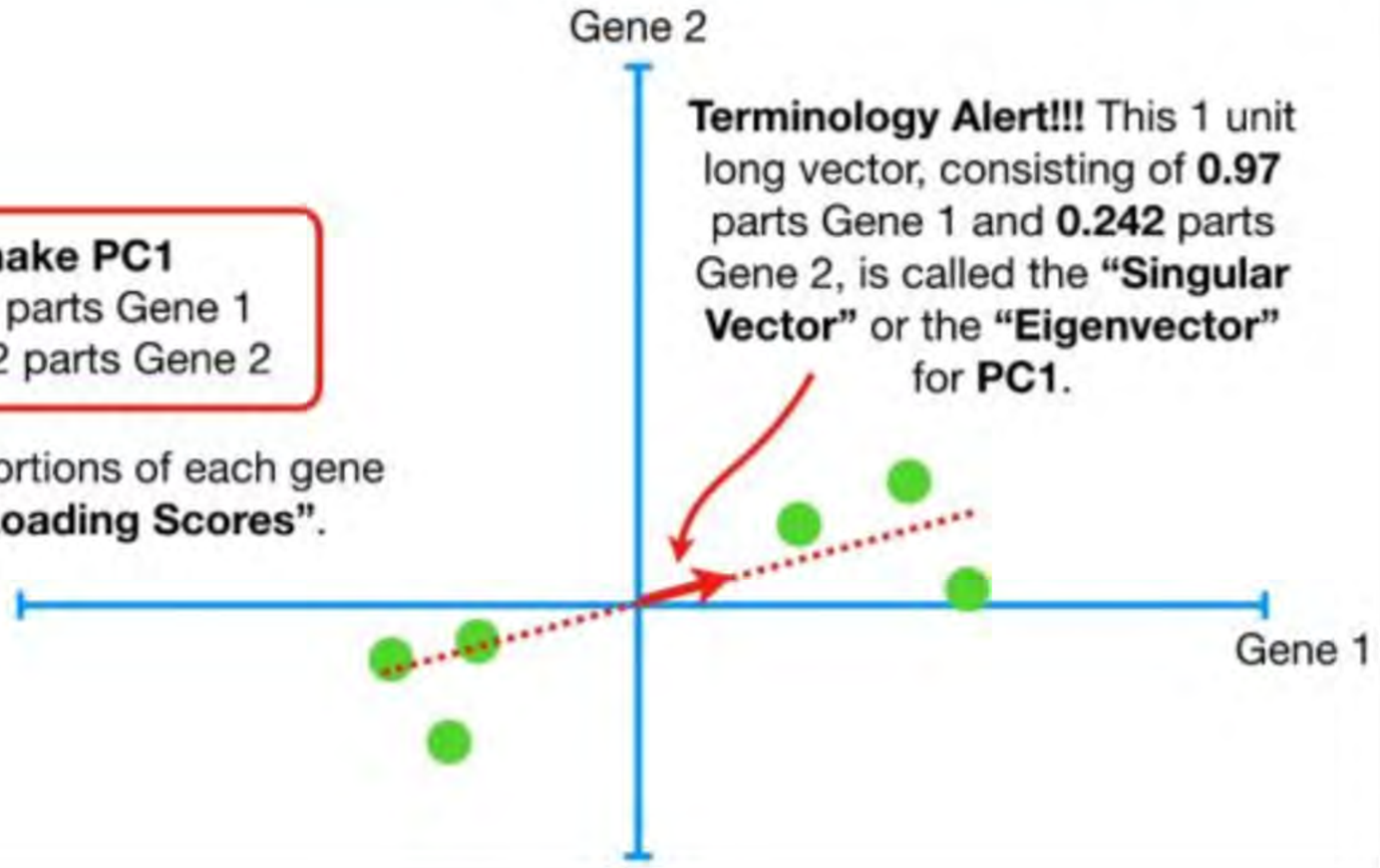
Terminology Alert!!!!
Mathematicians call this cocktail recipe a "*linear combination*" of Genes 1 and 2.



What is PCA

To make PC1
Mix 0.97 parts Gene 1
with 0.242 parts Gene 2

…and the proportions of each gene are called **"Loading Scores"**.



Terminology Alert!!! This 1 unit long vector, consisting of **0.97** parts Gene 1 and **0.242** parts Gene 2, is called the **"Singular Vector"** or the **"Eigenvector"** for **PC1**.

What is PCA

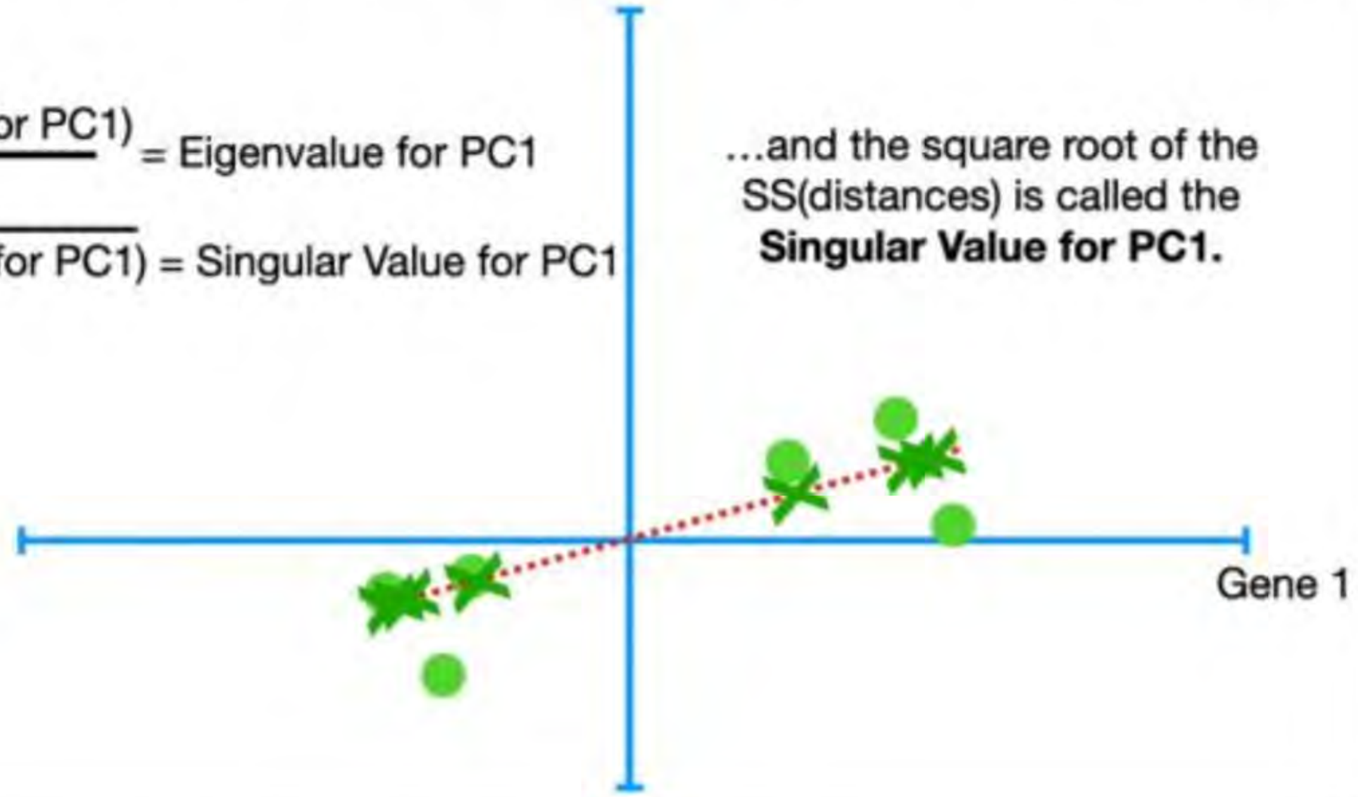
What this eigen value shows …

$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

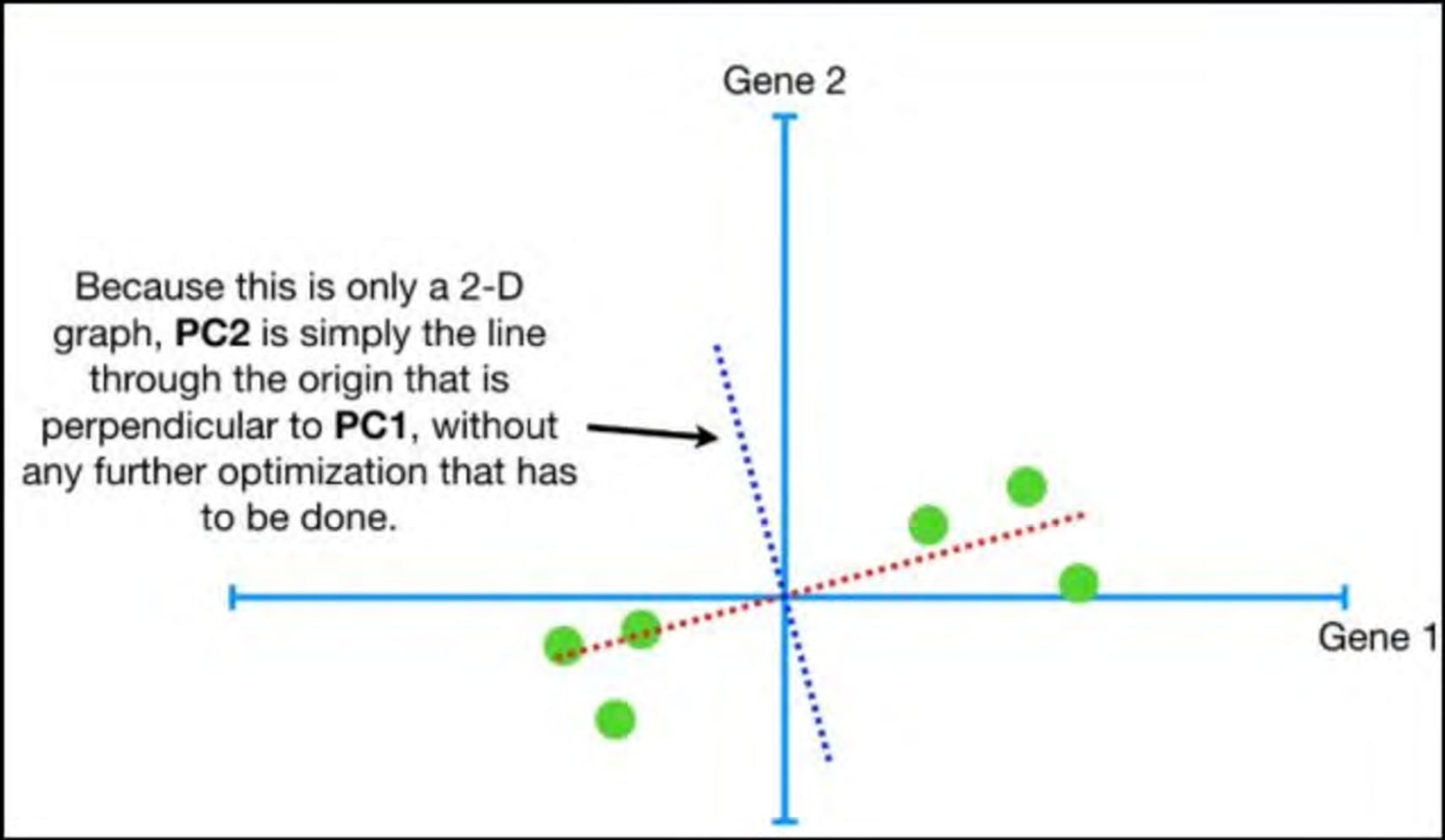
$$\frac{\text{SS(distances for PC1)}}{n - 1} = \text{Eigenvalue for PC1}$$

$$\sqrt{\text{SS(distances for PC1)}} = \text{Singular Value for PC1}$$

…and the square root of the SS(distances) is called the **Singular Value for PC1.**



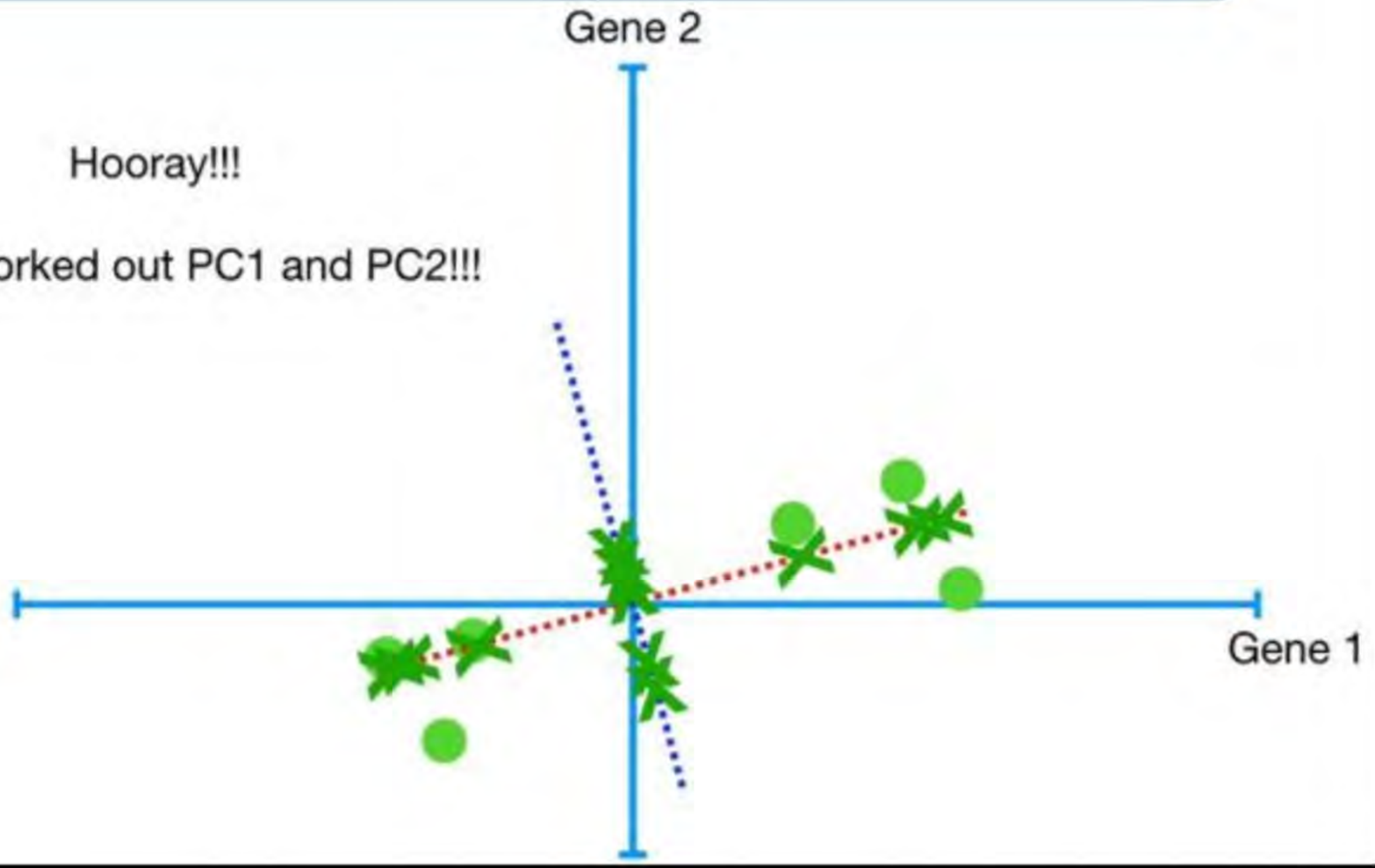
What is PCA


Gene 2
Because this is only a 2-D graph, **PC2** is simply the line through the origin that is perpendicular to **PC1**, without any further optimization that has to be done.
Gene 1

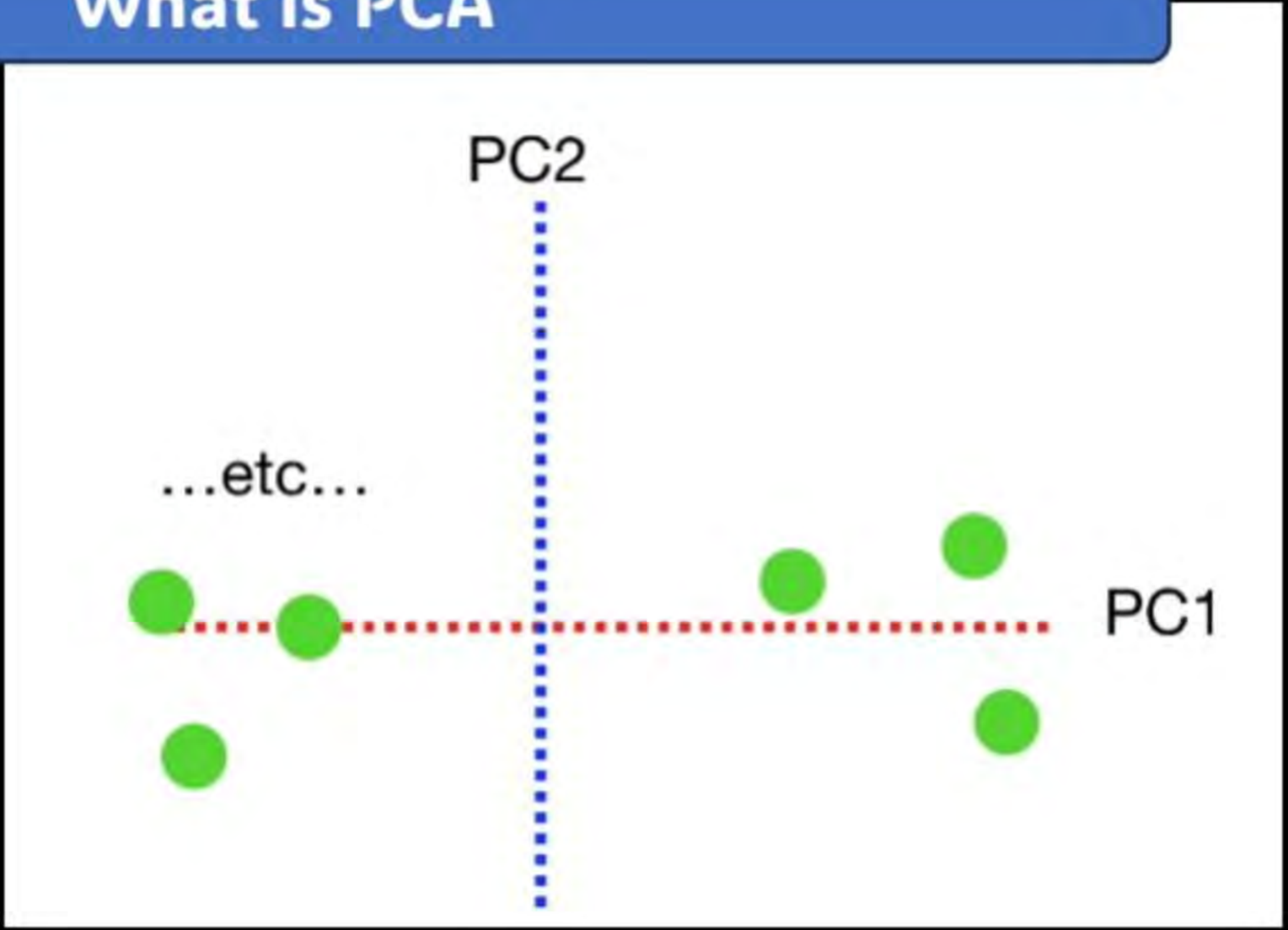
What is PCA

Hooray!!!

We've worked out PC1 and PC2!!!



What is PCA



THANK - YOU